

NAZARBAYEV INTELLECTUAL SCHOOL OF PHYSICS AND MATHEMATICS IN AKTOBE

Yearly course calendar for chemistry 2023-2024 academic year

Subject: Chemistry

Grade: 11

Language of instruction: English

Hours: 216

Weeks: 36

Units	Theme/Contents	Learning objectives
Term 1		
11.1A Atomic structure	• Fundamental particles (p,n,e) • Mass number, atomic (proton) number • isotopes • Relative atomic (molecular, formula) mass • The mass spectrometer • Electronic arrangement • Atomic orbitals and the Aufbau principle • Ionisation energy and the factors affecting it	11.1.1.1 know fundamental particles of atom 11.1.1.2 understand the meaning of mass number, atomic (proton) number 11.1.1.3 be able to write and work with full atomic symbols 11.1.1.4 be able to calculate relative atomic, molecular and formula mass 11.1.2.1 understand why isotopes occur and the nature of radioactivity 11.1.2.2 recall a block diagram of the mass spectrometer 11.1.2.3 understand the function of the main parts of the mass spectrometer 11.1.2.4 be able to interpret a simple mass spectrum to obtain relative atomic mass and relative molecular mass 11.1.3.1 understand and be able to work with a shell model of the atom: shell, sub-shell, orbital 11.1.3.2 recall the shapes of s, p, d, and f orbital (sets) 11.1.3.3 understand the rules for the filling of shells and sub-shells 11.1.3.4 recall the Aufbau (Kletchkovsky) principle as a mnemonic for the arrangement of electrons 11.1.3.5 be able to draw the electronic configuration for the first 36 elements 11.1.3.6 understand ionisation energy 11.1.3.7 understand the factors affecting ionisation energy: nuclear charge, distance from the nucleus, shielding
11.1B Amount of substance	• Relative atomic, molecular and formula mass • The Avogadro Constant and the mole • Equations (full and ionic, NOT half) • Calculation of reacting quantities and volumes (of gases), including titration and aqueous solutions • Empirical and molecular formula	11.1.1.5 understand the Avogadro number and mole (of particles) 11.2.2.1 be able to carry out calculations involving quantities of substances expressed in moles 11.2.2.2 be able express solution concentrations in mol dm ⁻³ 11.2.2.3 recall and understand the use of the 'molar volume' 11.2.2.4 recall and be able to use the ideal gas equation 11.2.2.5 know how to balance equations 11.2.2.6 know how to balance ionic equations 11.2.2.7 understand the purpose of, be able to carry out, and be able to carry out calculations involving, titration

		11.1.1.6 understand the use of empirical and molecular formulas 11.1.1.7 be able to calculate empirical and molecular formulas from analysis data 11.2.2.8 be able to calculate theoretical yield and percentage yield of reactions 11.2.2.9 understand and be able to calculate atom economy
11.1C Bonding	• Metallic, ionic, covalent and co-ordinate bonding • Crystal structures • Electronegativity and bond polarity • Intermolecular forces (dipole-dipole, van der Waals, hydrogen bonding)	11.1.4.1 understand the formation of ions and the charges which result 11.1.4.2 understand the formation of the ionic bond by electrostatic attraction 11.1.4.3 be able to deduce formulas of ionic compounds by a zero-sum method 11.1.4.4 understand how the bond type leads to ionic lattice structures 11.1.4.5 understand how the ionic lattice leads to the observed properties of ionic compounds 11.1.4.6 understand the formation of bonds by the sharing of electrons 11.1.4.7 understand the bonding in homonuclear diatomic molecules 11.1.4.8 recognise and understand the formation of double and triple bonds 11.1.4.9 understand the formation of macromolecular covalent lattices 11.1.4.10 understand the properties of ionic and covalent lattice structures 11.1.4.11 be able to draw 'dots and crosses' diagrams to show the formation of covalent compounds 11.1.4.12 understand the nature of a metallic lattice 11.1.4.13 understand the nature of the metallic bond as enabled by delocalised electrons 11.1.4.14 be able to draw a simple representation of a metallic lattice and the metallic bond 11.1.4.15 understand how metallic bonding leads to the observed properties of metals 11.1.4.16 be able to compare and contrast the properties and structures associated with the various bond types 11.1.4.17 understand the concept of electronegativity/ electropositivity 11.1.4.18 understand the factors which affect electronegativity 11.1.4.19 understand bond polarity, its causes and effects 11.1.4.20 understand the causes and effects of dipole-dipole forces 11.1.4.21 understand the cause and effects of van der Waals (induced dipole-dipole) forces 11.1.4.22 understand the cause and effects of hydrogen bonding 11.1.4.23 be able to compare and contrast the three types of force and their effects:

		dipole-dipole, van der Waals and the forces in hydrogen bonds 11.1.4.24 recognize electron pairs as bond pairs or lone pairs 11.1.4.25 recognize and understand the occurrence of dative covalent (co-ordinate) bonds 11.1.4.26 be able to use valence shell electron pair repulsion theory (VSEPR) to predict the shapes of molecules 11.1.4.27 be able to draw representations of molecular shapes
11.1D Periodicity	• The Aufbau principle and the Periodic table • Classify elements as s,p,d block • Trends in periods and groups • Elements of period 3 (Na - Ar) • Acid/base properties of period 3	11.2.1.1 recognize a number of forms of the Periodic Table, historic and modern 11.2.1.2 be able to use the Aufbau Principle to explain the shape of the modern Periodic Table 11.2.1.3 understand the significance of the s, p, d and f blocks 11.2.1.4 be able to use the table to read and write the electronic configuration of any element 11.2.1.5 recognize periodicity in the table and be able to account for this in terms of effective nuclear charge, ionisation energy, bond types etc. 11.2.1.6 recall and be able to explain general trends in melting and boiling points, atomic radius, first ionisation energy, acid/base properties (recall and explain general trends using the elements Na to Ar as exemplification) 11.2.1.7 recognise and be able to account for, in terms of effective nuclear charge, the following trends down groups: bond types, melting and boiling points, reactivity etc. (recognise and be able to account for trends down groups using (as a minimum) groups 1 and 17 as exemplification)
Term 2		
11.2A Introduction to organic chemistry - alkanes	• Formula representations: empirical, molecular, structural, displayed • Functional groups • Homologous series • IUPAC naming conventions: aliphatic compounds only • Chain, positional, functional, stereoisomerisms • Hybrid orbitals, sigma and pi bonds	11.4.2.1 recognise and understand the use of a variety of formula types 11.4.2.2 understand that an organic compound generally consists of a hydrocarbon component plus a functional group 11.4.2.3 recognise the common functional groups 11.4.2.4 understand and be able to use the IUPAC nomenclature 11.4.2.5 understand the formation of homologous series and the similarity of properties 11.4.2.6 understand why isomerism occurs and be able to draw isomers for given molecular formulas 11.4.2.7 understand the formation of sigma and pi bonds and know their occurrence

	• Alkanes as basic structures • Petroleum: fractionation and cracking • Combustion products and environmental considerations • Halogenation	and the implication for molecular shapes 11.4.2.8 understand the concept of saturation and apply it to the alkanes as a 'base' homologous series 11.4.2.9 recognise the importance of alkanes as fuels, their combustion products and the environmental effects of burning hydrocarbon fuels 11.4.2.10 recall details of the fractionation of oil and the uses of the fractions 11.4.2.11 understand homolytic bond fission and the formation of free radicals 11.4.2.12 understand catalytic cracking and thermal cracking in terms of a free radical process 11.4.2.13 know and understand the mechanism of photochemical chlorination
11.2B Energetics	• Enthalpy and enthalpy change • Bond enthalpies • Calorimetry • Hess's law, including simple applications	11.3.1.1 understand and be able to define enthalpy and enthalpy change (ΔH) 11.3.1.2 understand the need for standard enthalpies and be able to define standard enthalpy 11.3.1.3 understand that chemical reactions involve the making and breaking of bonds and the concept of bond enthalpy 11.3.1.4 be able to determine bond enthalpies from data 11.3.1.5 be able to use mean bond enthalpies to estimate ΔH for reactions 11.3.1.6 be able to determine simple enthalpy changes experimentally and calculate enthalpy change from experimental data 11.3.1.7 Know definition and be able to use it calculate enthalpy changes in reactions
11.2C Kinetics	• States of matter • The ideal gas equation • Collision theory and activation energy • Maxwell-Boltzmann Distribution • Effect of temperature • Effect of concentration • Rate equations	11.3.2.1 recall and be able to apply the ideas from earlier work on the states of matter and simple collision theory 11.3.2.2 be able to apply the ideal gas equation and the molar volume to reactions involving gases 11.3.2.3 understand the importance of energy in gas and liquid collisions: the Maxwell Boltzmann distribution and activation energy 11.3.2.4 use these concepts to give a comprehensive account of the effects of temperature and concentration on the rate of a reaction

	• Determination of a rate equation • Catalysts	11.3.2.5 understand that the rate of a chemical process can be represented by a rate equation 11.3.2.6 recall the general rate equation and understand the rate constant and the order of reaction 11.3.2.7 be able to investigate how the rate of a reaction changes with changing conditions 11.3.2.8 be able to carry out simple calculations of rates and rate constants 11.3.2.9 know the meaning and definition of ‘catalyst’ 11.3.2.10 understand the process of catalysis and how catalysis works 11.3.2.11 distinguish between homogeneous and heterogeneous catalysts 11.3.2.12 be able to describe the process of heterogeneous catalysis using a particle explanation
11.2D Equilibria	• Dynamic equilibria • Effects of change of pressure, temperature, concentration (qualitative) • Industrial equilibria • Equilibrium constant K_c • Effect of concentration and temperature on K_c (qualitative)	11.3.3.1 recall and be able to use ideas from previous teaching on equilibria, including the concept of dynamic equilibrium 11.3.3.2 be able to predict the effects change of temperature, concentration and pressure on equilibria 11.3.3.3 understand why a catalyst does not affect the position of equilibrium 11.3.3.4 understand the importance of equilibria in industrial chemistry 11.3.3.5 understand (with examples) why reaction conditions for industrial processes involving equilibria are often a compromise 11.3.3.6 be able to write the equilibrium constant (K_c) for a given reaction 11.3.3.7 be able to predict the effect of changing temperature on K_c and understand that K_c is not affected by changes in concentration or the addition of a catalyst
Term 3		
11.3A Alkenes	• Structure, bonding and reactivity • Stereoisomerism (cis-trans, E-Z) • Addition reactions • Polymerisation	11.4.2.14 understand the term ‘unsaturation’ and its consequences for the compound 11.4.2.15 recall tests for unsaturation 11.4.2.16 recognise the possibilities of geometric isomerism (‘E-Z’ or cis-trans isomerism) and recognise it as a type of stereoisomerism 11.4.2.17 understand the terms ‘electrophile’ and ‘nucleophile’ and be able to classify reagents 11.4.2.18 understand the mechanism of electrophilic addition, know some examples and be able to apply the mechanism to new cases 11.4.2.19 be able to predict the products of addition to unsymmetrical alkenes

		11.4.2.20 know some industrially important addition reactions 11.4.2.21 understand addition polymerisation 11.4.2.22 understand at a molecular level why addition polymers are relatively inert 11.4.2.23 know the outline of manufacturing processes for (poly)alkenes 11.4.2.24 know some common uses of (poly)alkenes and understand the need for recycling
11.3B Alcohols	• Classification • Reactions: substitution and elimination • Industrial manufacture of ethanol	11.4.2.25 understand and be able to apply the classification of alcohols 11.4.2.26 understand the oxidation of alcohols and their products 11.4.2.27 understand the products of dehydration of alcohols 11.4.2.28 understand the two main routes for ethanol production and the advantages and disadvantages of each 11.4.2.29 understand the environmental issues of ethanol production including those of bioethanol fuels, and understand the concept of carbon neutrality 11.4.2.30 be able to discuss biofuels in the context of environmental issues
11.3C Haloalkanes	• Reactions: • Photo chlorination (free radical) • Nucleophilic substitution • Elimination	11.4.2.31 recall the synthesis of chloroalkanes 11.4.2.32 understand environmental concerns about haloalkanes and understand the mechanism of ozone depletion 11.4.2.33 know less environmentally damaging substitutes for haloalkanes 11.4.2.34 recognise that haloalkanes will react with nucleophiles 11.4.2.35 understand the mechanism of nucleophilic substitution reactions 11.4.2.36 be able to write equations and mechanisms for a general case, and some common examples of, nucleophilic substitution 11.4.2.37 understand elimination and its mechanism 11.4.2.38 understand the competition between substitution and elimination 11.4.2.39 understand the importance of haloalkanes as intermediates in synthesis (including industrial processes)
11.3D Analytical techniques	• Gas-liquid chromatography	11.1.4.28 understand the importance of analytical techniques in modern chemistry 11.1.4.29 recall previous work on chromatography and understand gas-liquid chromatography as an extension of it

	• Mass spectrometry • Infra-red spectroscopy • NMR spectroscopy	11.1.4.30 understand the importance of gas-liquid chromatography as a precursor process in many forms of analysis 11.1.4.31 recall the block diagram and outline operation of the mass spectrometer 11.1.4.32 understand at an elementary level the production of the molecular ion and its subsequent fragmentation 11.1.4.33 know in outline the cause of the M+1 peak 11.1.4.34 be able to find relative molecular mass from a (simplified) mass spectrum 11.1.4.35 be able to make sensible suggestions about structure by inspection of the fragmentation pattern 11.1.4.36 understand the reasons for absorption of infrared frequencies by a molecule 11.1.4.37 understand and know the location of the 'fingerprint region' and be able to use it to identify compounds 11.1.4.38 understand the implications of infrared absorption for global warming 11.1.4.39 understand an outline of C-13 NMR and proton NMR and how spectra are obtained 11.1.4.40 understand the origin of, and be able to interpret a simple C-13 NMR spectrum 11.1.4.41 understand the origin of, and be able to interpret a simple proton NMR spectrum 11.1.4.42 relate this to the use of magnetic resonance imaging 11.1.4.43 understand that these techniques can yield information both by empirical analysis of the spectral data or by comparison with the spectra of known substances
Term 4		
11.4A Redox reactions	• Oxidation states • Redox equations • Electrode potentials • Electrochemical series • Electrochemical cells	11.2.3.1 recall concepts from earlier grades: oxidation as gain of oxygen and reduction as gain of hydrogen, oxidation and reduction in terms of electron gain and loss 11.2.3.2 know and be able to use the rules for assigning oxidation numbers and be able to assign oxidation numbers to individual elements in compounds 11.2.3.3 be able to write oxidising and reducing half equations and to combine them into an overall redox equation 11.2.3.4 use electrode potentials to establish an electrochemical series 11.2.3.5 understand the electrochemical cell in terms of conventional representations

		11.2.3.6 know that standard electrode potentials require standard conditions 11.2.3.7 understand how cells are used to measure electrode potentials 11.2.3.8 be able to use standard electrode potential to calculate the EMF of a cell and to predict the directions of redox reactions 11.2.3.9 understand electrochemical cells as a source of energy, including the constituents of commercial cells 11.2.3.10 understand the processes of charging and discharging cells 11.2.3.11 understand the construction and operation of a hydrogen-oxygen fuel cell 11.2.3.12 understand the benefits and potential risks of fuel cells
11.4B Group 17 (The Halogens)	• Trends in physical properties • Trends in oxidising ability • Trends in reducing ability • Identification of halide ions in aqueous solution • Uses of chlorine and its compounds	11.2.1.8 understand and be able to account for the trends in electronegativity and boiling point, be able to account for the decreasing oxidising ability down the group 11.2.1.9 recognise the displacement reactions of the halogens 11.2.1.10 understand and be able to account for the trend in reducing abilities of the halide ions 11.2.1.11 be able to identify halide ions using silver nitrate and ammonia solution 11.2.1.12 know how chlorine water is formed and the reactions of chlorine with aqueous sodium hydroxide 11.2.1.13 understand the use of chlorine as a water treatment and understand the balance of risks and benefits in this process
11.4C Group 2: The alkaline earth elements	• Trends in physical properties • Trends in chemical properties	11.2.1.14 understand and be able to account for trends in melting point, atomic radius, first ionisation energy, electronegativity (electropositivity) 11.2.1.15 know and be able to account for the trends in reactivity with water, the relative solubility of the hydroxides and sulphates 11.2.1.16 know and be able to explain the uses of Mg(OH)_2 and Ca(OH)_2 11.2.1.17 know the use of BaCl_2 solution to test for sulphates and the use of BaSO_4 in medicine

11.4D Thermodynamics	• Enthalpy change (ΔH) • Entropy change (ΔS) • Free energy change (ΔG) • Use of $\Delta G = \Delta H - T\Delta S$ to determine reaction feasibility	11.3.1.7 understand the underlying principle of Hess' Law and be able to use it calculate enthalpy changes in reactions 11.3.1.8 be able to calculate ΔH from bond enthalpies 11.3.1.9 be able to construct a Born-Haber cycle and compare lattice energies from this approach with those from other calculations 11.3.1.10 be able to define and use the concepts of free energy change (ΔG) and entropy change (ΔS) 11.3.1.11 understand that enthalpy change alone is not sufficient to explain spontaneous changes and recognise the role of entropy 11.3.1.12 know and be able to use the equation $\Delta G = \Delta H - T\Delta S$ to show how ΔG changes with temperature and to find the temperature at which a reaction will proceed
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